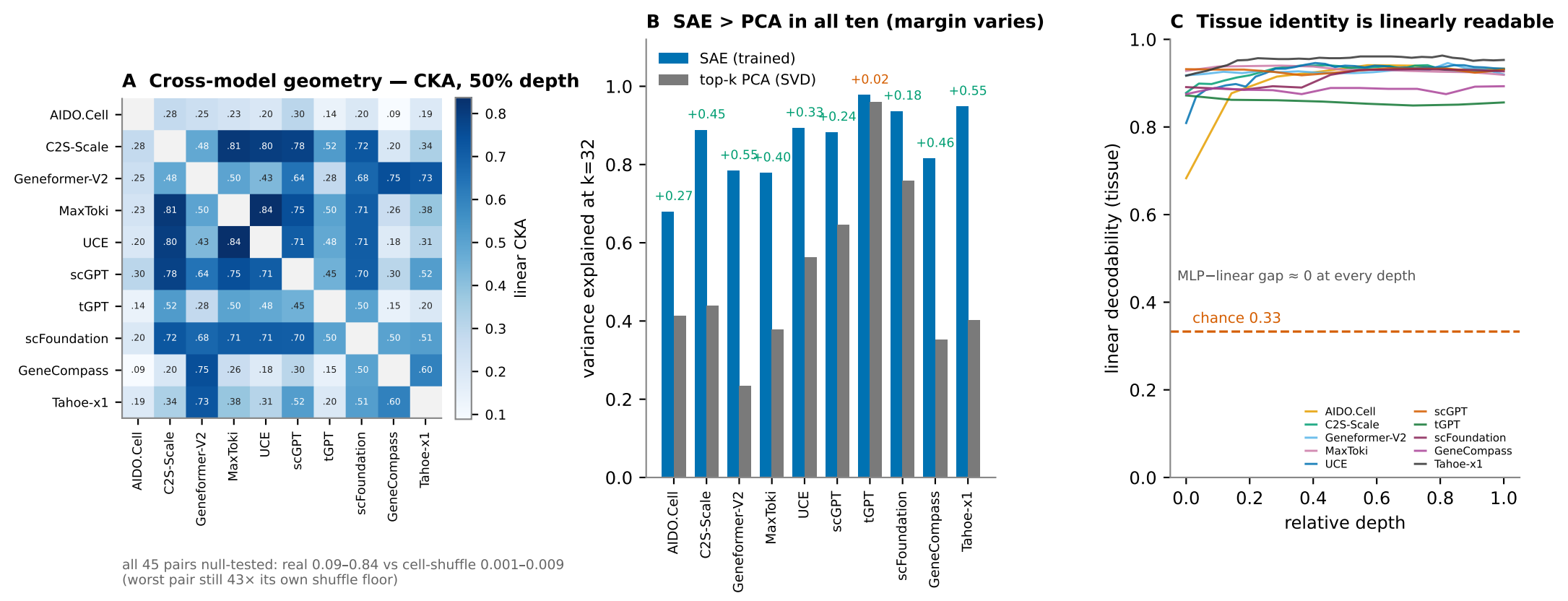
A Cross-model geometry — CKA, 50% depth
AIDO.Cell .28 .25 .23 .20 .30 .14 .20 .09 .19
C2S-Scale .28 .48 .81 .80 .78 .52 .72 .20 .34
Geneformer-V2 .25 .48 .50 .43 .64 .28 .68 .75 .73
MaxToki .23 .81 .50 .84 .75 .50 .71 .26 .38
UCE .20 .80 .43 .84 .71 .48 .71 .18 .31
scGPT .30 .78 .64 .75 .71 .45 .70 .30 .52
tGPT .14 .52 .28 .50 .48 .45 .50 .15 .20
scFoundation .20 .72 .68 .71 .71 .70 .50 .50 .51
GeneCompass .09 .20 .75 .26 .18 .30 .15 .50 .60
Tahoe-x1 .19 .34 .73 .38 .31 .52 .20 .51 .60
AIDO.Cell C2S-Scale Geneformer-V2 MaxToki UCE scGPT tGPT scFoundation GeneCompass Tahoe-x1
linear CKA
0.8 0.7 0.6 0.5 0.4 0.3 0.2 0.1
all 45 pairs null-tested: real 0.09–0.84 vs cell-shuffle 0.001–0.009
(worst pair still 43× its own shuffle floor)
B SAE > PCA in all ten (margin varies)
SAE (trained)
top-k PCA (SVD)
variance explained at k=32
1.0 0.8 0.6 0.4 0.2 0.0
+0.27 +0.45 +0.55 +0.40 +0.33 +0.24 +0.02 +0.18 +0.46 +0.55
AIDO.Cell C2S-Scale Geneformer-V2 MaxToki UCE scGPT tGPT scFoundation GeneCompass Tahoe-x1
C Tissue identity is linearly readable
linear decodability (tissue)
1.0 0.8 0.6 0.4 0.2 0.0
MLP−linear gap ≈ 0 at every depth
chance 0.33
AIDO.Cell
C2S-Scale
Geneformer-V2
MaxToki
UCE
scGPT
tGPT
scFoundation
GeneCompass
Tahoe-x1
0.0 0.2 0.4 0.6 0.8 1.0
relative depth